INCH-POUND

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2 January 2009
SUPERSEDING
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5 October 1982

DETAIL SPECIFICATION

SPOTLIGHTS, INCANDESCENT, ROOF-MOUNTED, AUTOMOTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers roof-mounted spotlights of 6, 12, and 24-volt ratings, using PAR 46, screw-terminal lamps of the proper voltage and power for use on commercial vehicles.

1.2 Classification. The spotlights covered in this specification shall be of the following types, classes, and sizes as specified below (see 6.2):

1.2.1 Types. The types of spotlights are as follows:

Type I - Ready for use after mounting with no further treatment
Type II - Primed to receive paint matching vehicle paint

1.2.2 Classes. The classes of spotlights are as follows:

Class A - 6-inch control arm
Class B - 23-inch control arm

1.2.3 Sizes. The sizes of spotlights are detailed in table I.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil/.

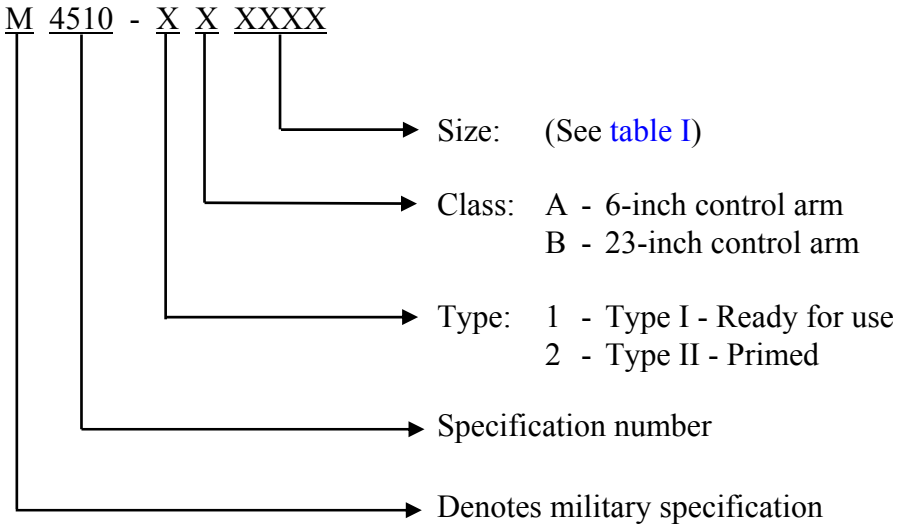
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TABLE I. Size, sealed beam trade numbers and operating characteristics.

Size	Trade no.	Design volts	Watts	Nominal C.P.	Beam horiz	Spread vert	Rated life[1] hours	Government part number
4535	4535	6.4	30	95,000	5.5°	4°	100	A52409-4535
4533	4533	6.5	40	50,000	7°	7°	50	A52409-4533
4435	4435	12.8	30	75,000	5°	5°	100	A52409-4435
4530	4530	26.0	135	100,000	11°	11°	50	A52409-4530
4553	4553	28.0	250	225,000	2°	2°	25	MS25241-4553

[1] At design volts.

1.3 Part or identifying number (PIN). The PIN to be used for spotlights acquired to this specification is created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloy
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-889	- Dissimilar Metals

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	- General Guidelines for Electronic Equipment
MIL-HDBK-808	- Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support and Equipment

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

IPC - ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES

J-STD-006	- Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications

(Copies of this document are available online at http://www.ipc.org/ or from Institute for Interconnecting and Packaging Electronic Circuits, 2215 Sanders Road, Northbrook, IL 60062-6135.)

SAE INTERNATIONAL

SAE AMS-QQ-P-416	- Plating, Cadmium (Electrodeposited)
SAE J575	- Test Methods and Equipment for Lighting Devices and Components for Use on Vehicles Less than 2032 mm in Overall Width

(Copies of these documents are available online at http://www.sae.org/ or from the SAE International, 400 Commonwealth Drive, Warrendale, PA 05906.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials shall be the supplier's standard commercial material with such changes as are necessary to meet the requirements of this specification.

3.2.1 Solder. Solder employed in making electrical connections shall be a rosin core type conforming to compositions Sn50Pb50, Sn60Pb40 or Sn70Pb30 of J-STD-006.

3.2.2 Metal. The lamp housing shall be fabricated of metal.

3.2.2.1 Dissimilar metals. Dissimilar metals shall be treated as specified in 3.3.2.

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. The spotlight shall be the manufacturer's standard item, modified as required to meet the requirements of this specification.

3.3.1 Soldering. Soldering shall be in accordance with MIL-HDBK-454.

3.3.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals shall be avoided. Where such contact is unavoidable, metal shall be treated to prevent deterioration by chemical or electrochemical reaction with its environment. Requirements for protecting coupled dissimilar metals shall be in accordance with MIL-STD-889.

3.3.3 Movement. The movement of the lamp housing shall be controlled by one hand operation of the control handle (knobs). The lamp housing shall move in a zenith thru an arc of not less than 270 degrees and shall move in elevation from vertical to not less than 35 degrees below horizontal.

3.3.4 Bulb. The spotlight shall be designed for insertion of bulbs of the PAR 46, screw terminal configuration. The bulb specified (see table I and 6.2) shall determine the size of the spotlight housing.

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3.4 Performance. The spotlight shall be capable of operating satisfactorily under the following conditions:

a. Temperature ranging from -54 °C to +71 °C.
b. Exposure to salt-sea atmosphere.
c. Vibration incident to service use.
d. Fungus growth as encountered in humid climates.
e. Rainfall as encountered in any locale.
f. Sand and dust conditions as encountered in arid locales.

3.5 Components. The manufacturer's standard commercial components shall be used with such modifications as are necessary to meet the dimensional and performance requirements of this specification. The spotlight shall consist of the following major components:

a. Lamp housing (see 3.5.1).
b. Lamp (see 3.5.2).
c. Support arm (see 3.5.3).
d. Control arm and handle (see 1.2.2 and 3.5.4).
e. Switch and wiring (see 3.5.5).
f. Roof seal (see 3.5.6).

3.5.1 Lamp housing. The lamp housing shall be fabricated of metal and shall be designed to hold and position PAR 46 screw terminal lamps. The indexing notch for the lamp shall be positioned to locate the widest divergence of the beam in the horizontal plane. The housing's diameter shall be not greater than 6.3125 inches. The housing's length shall be not greater than 6 inches.

3.5.2 Lamp. The lamp shall be a PAR 46, screw terminal base specified in table I.

3.5.3 Support arm. A support arm with a mounting flange suitable for mounting the spotlight on a plane surface shall be provided. The support arm shall support and position the lamp head under all conditions specified herein. The length of the arm shall be such that the distance from the horizontal centerline of the lamp housing to the face of the mounting flange shall be 6.5 inches ± 0.25 inch. Mounting screws or bolts of sufficient length to fasten the arm and adapter (see 3.5.6) to a panel or bracket 0.125 inch in thickness shall be furnished complete with self-locking nuts or nuts and lock washers.

3.5.4 Control arm and handle. A control arm and handle as specified in 1.2.2 and 6.2 shall be furnished. The arm shall be an integral part of or fastened to the support arm and flange (3.5.3). The control arm, when coupled to the support arm shall completely enclose all moving parts. The length of the control arm (see 1.2.2) shall be ± 0.125 inch measured from the mounting surface of the mounting flange (see 3.5.3) to the horizontal centerline of the control handle or knob. The diameter of control knobs shall be not greater than 4 inches. Control handles shall not extend more than 5 inches from the vertical centerline of the control arm. The operations of the control knob or handle shall give the movement specified in 3.3.3.

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3.5.5 Switch and wiring. The switch and electrical conductors shall be furnished to meet electrical voltage and current requirements specified for the particular lamp size from table I. The switch shall be incorporated into the control handle or a bracket mounted on the control arm. Brackets, when furnished, shall have the ability to be positioned at least 270 degrees about the arm. Provisions shall be made to permit electrical power connection to the spotlight after the installation is completed. Wires shall be of a length to permit bulb installation and connection with the bulb rotated 90 degrees from its proper installed position (see 3.5.1).

3.5.6 Roof seal. A sealing gasket to fill the voids remaining when the mounting flange contacts a convex surface as found on commercial vehicle roofs shall be furnished.

3.6 Lubrication. All internal moving parts shall be lubricated prior to assembly. All openings shall be sealed to prevent leakage. The light shall not require subsequent lubrication.

3.7 Finishes and protective coatings. All finishes and protective coatings shall be in accordance with MIL-HDBK-808.

3.7.1 Type II. Type II lights shall be prime painted on the exposed surfaces of exterior assemblies, lamp head, support arm, etc. Other treatment shall be as type I.

3.7.2 Type I. Type I spotlights and interior parts of type II spotlights shall be furnished as follows:

a. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes need not be anodized.

b. Aluminum alloy which are not subject to abrasion and which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-DTL-5541.

c. Ferrous parts, except those fabricated from corrosion-resistant steel, shall be cadmium plated in accordance with SAE AMS-QQ-P-416.

d. All parts normally painted or plated in commercial practice shall be treated in accordance with the manufacturer's standard practice if not in conflict with 3.7.1 or the balance of this paragraph.

3.8 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8.1 Nameplate. A nameplate shall be permanently fastened to the lamp housing approximately 180 degrees from the mounting, containing the following information:

a. Spotlight, Roof-Mounted.
b. Max. Volts.
c. Max. Amps.
d. Manufacturer's Model Number.

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e. Manufacturer's Name.
f. US.
g. Military Part Number.

3.9 Instructions. Mounting instructions, including required templates shall be packed with each light.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.2 and 4.3 of this specification.

4.2.1 Lamp failure. Failed lamps shall not constitute light failure. The lamp shall be replaced and the tests continued.

4.3 First article inspection. A sample of the spotlight the supplier proposes to furnish under the specification shall be examined and tested as follows:

4.3.1 Conformance requirement. The light shall be examined to assure conformance to all requirements of section 3 of this specification.

4.3.2 Operation. The light shall be operated thru the full angular range specified in 3.3.3 with the lamp on. Conformance to the bulb position specified in 3.5.1 shall be verified. No binding or looseness of parts shall be evident. The lamp head shall retain its position when the control handle is released.

4.3.3 Environmental inspection. Unless otherwise specified (see 6.2), the light with lamp installed shall be subjected to the following tests:

4.3.3.1 High temperature. The light shall be placed in a test chamber having a temperature of +71 ºC ± 1º, with a relative humidity of not more that 15 percent. After a soaking period of not less than 24 hours the light shall be operated in accordance with paragraph 4.3.2. The light shall then be allowed to return to normal ambient conditions and shall be examined in accordance with 4.3.1 and operated in accordance with 4.3.2.

4.3.3.2 Low temperature. The light shall be placed in a test chamber having a temperature of -54 ºC. After a soaking period of not less than 24 hours, the spotlight shall be operated in accordance with 4.3.2, first with the lamp off, then with the lamp on. The light shall be allowed to return to normal ambient conditions and examined in accordance with 4.3.2.

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4.3.3.3 Vibration. The spotlight shall be subjected to vibration in accordance with the vibration test of SAE J575. Upon completion of the test, the light shall be examined in accordance with 4.3.1 and operated in accordance with 4.3.2.

4.3.3.4 Moisture. The spotlight shall be subjected to moisture in accordance with the moisture test of SAE J575. Upon completion of this test, the light shall be operated in accordance with 4.3.2.

4.3.3.5 Dust. The spotlight shall be subjected to dust in accordance with the dust test of SAE J575. Upon completion of this test, the light shall be operated in accordance with 4.3.2.

4.3.3.6 Corrosion. The spotlight shall be subjected to corrosion in accordance with the corrosion test of SAE J575. Upon completion of this test, the spotlight shall be operated in accordance with 4.3.2.

4.3.4 Ease of maintenance. Such components as the sealed beam unit, control handle, switch, gears, adapter, and wiring shall be removed and reinstalled and those adjustments necessary shall be made to determine ease of maintenance without special tools.

4.4 Conformance inspection. Each spotlight shall be subjected to the tests in 4.3.1 and 4.3.2 before acceptance.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The spotlights covered by this specification are intended for use in the following applications:

a. Class A spotlights are intended for universal use including driver operated mountings.

b. Class B spotlights are intended for use from stationary vehicles and are not to be cab roof mounted.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class and size required (see 1.2).
c. When first article inspection is required (see 3.1).
d. Whether environmental inspections are not required (see 4.3.3).
e. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Bulb
Control arm
Housing, lamp
Lamp, sealed beam
Metals, dissimilar
Seal, roof
Solder
Soldering

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99

Preparing Activity:
DLA - GS2
(Project 6220-2009-001)

Review Activity:
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